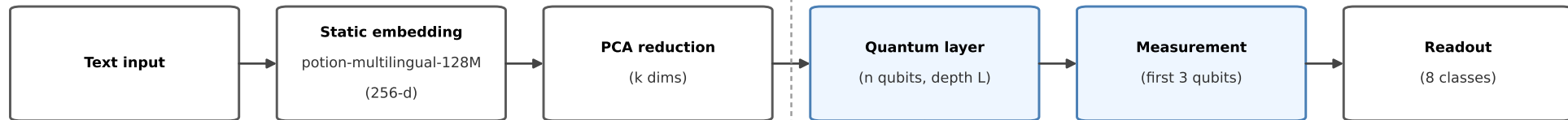


classical front end quantum layer (simulated)



Cross-validation protocol

Every circuit is evaluated twice:

- CUDA-Q (formal track; the reported numbers)
- an independent NumPy state-vector simulator

Agreement criterion: $\max |\Delta P| < 1e-10$.

Both tracks share one golden-vector test set,
so a silent divergence cannot pass unnoticed.

